CNG 351 - Assignment 1

Data Management and File Structures

Fall 2026-2027

Entity-Relationship Database Designs

Team / Group ID: 8

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Notation and model organization

The diagrams use eight views: 1A-1B for Use Case 1, 2A-2C for Use Case 2 and 3A-3C for Use Case 3. When an entity appears in more than one view, it is the same entity, with the same key, attributes and relationships.

Chen notation

Symbol / marking	Meaning
Single rectangle	Strong entity type.
Double rectangle	Weak entity; its full identity includes its owner's key.
Single / double diamond	Binary relationship / identifying relationship.
Oval / branched oval	Simple attribute / composite attribute with components.
Double oval / dashed oval	Multivalued attribute / derived attribute.
Solid underline	Chosen entity key. Other uniqueness constraints are stated in the rules.
Dashed underline	Partial key of a weak entity. BookingDate and StartTime together form Booking's partial key.
Single / double connection	Partial participation / total participation of that entity in the relationship.
(min,max) beside an entity	Number of relationship occurrences for each instance of that entity; N means many.

How to read a constraint

In Season (0,10) - Contains - Episode (1,1), the (0,10) allows a season to have zero to ten episodes. The (1,1) requires each episode to belong to exactly one season. Episode has a double line because its participation is total.

Additional rules

Each relationship diamond connects two entity roles. An attribute attached to a relationship does not add another role. Date restrictions, subset conditions and calculations that cannot be fully shown in the diagrams are written in the rules. EmployeeCount* is calculated automatically and stored.

Use Case 1

Who Wants to Be a Millionaire?

The quiz show is organised into seasons and episodes. QuestionAttempt records the questions asked during a contestant's game, and JokerUse records the jokers they use.

Entity identities

Entity	Chosen full key
Season	SeasonID
Episode	EpisodeID
Host	HostID
Contestant	ContestantID
Question	QuestionID
Category	CategoryID
QuestionAttempt (weak)	(ContestantID, AttemptNo)
JokerUse (weak)	(ContestantID, UseNo)
JokerType	TypeName

Design decisions

QuestionAttempt is weak under Contestant. AttemptNo identifies an attempt within that contestant's game and gives the question order. UsesQuestion links the attempt to its question.

JokerUse is also weak under Contestant. UsedFor links a use to an attempt, and OfType gives its joker type. These are binary relationships; the limit of four uses comes from Contestant's (0,4) participation in HasJokerUse.

ChoiceA, ChoiceB, ChoiceC and ChoiceD hold the four answers. CorrectChoice gives the label of the correct answer. A question can belong to several categories, but it must belong to at least one.

For both Host and Contestant, FullName is made up of FirstName and Surname. Contestant.Age is derived from DateOfBirth and counts completed years.

Use Case 1 - Rules and assumptions

Business rules and additional constraints

1. A season can contain up to ten episodes. Each episode has one season and one host. Each contestant belongs to one episode. We keep one Contestant record per person because a person can take part in the show only once.

2. Host.Email is unique as well as HostID. EpisodeNumber is stored separately; the brief does not require it to be globally unique.

3. A question has its text, four answer slots labelled A, B, C and D, and one CorrectChoice from these labels. DifficultyLevel holds a label such as easy, medium or hard. Every question belongs to at least one category.

4. A completed game includes at least one QuestionAttempt. AttemptNo is unique within a contestant's game and increases in question order. GivenAnswer holds the submitted answer label. IsCorrect is stored and must agree with GivenAnswer = Question.CorrectChoice.

5. A contestant can have zero to four JokerUse records. Each use records UsedAt, including the date and time, and links to one joker type and one QuestionAttempt. The attempt and the joker use must belong to the same contestant.

6. JokerType lists Fifty-Fifty, Phone a Friend and Ask the Audience, with a description for each.

Assumptions

A1. Seasons, planned episodes, hosts, categories and questions can be entered before they are used. When the brief gives no minimum participation, we use zero.

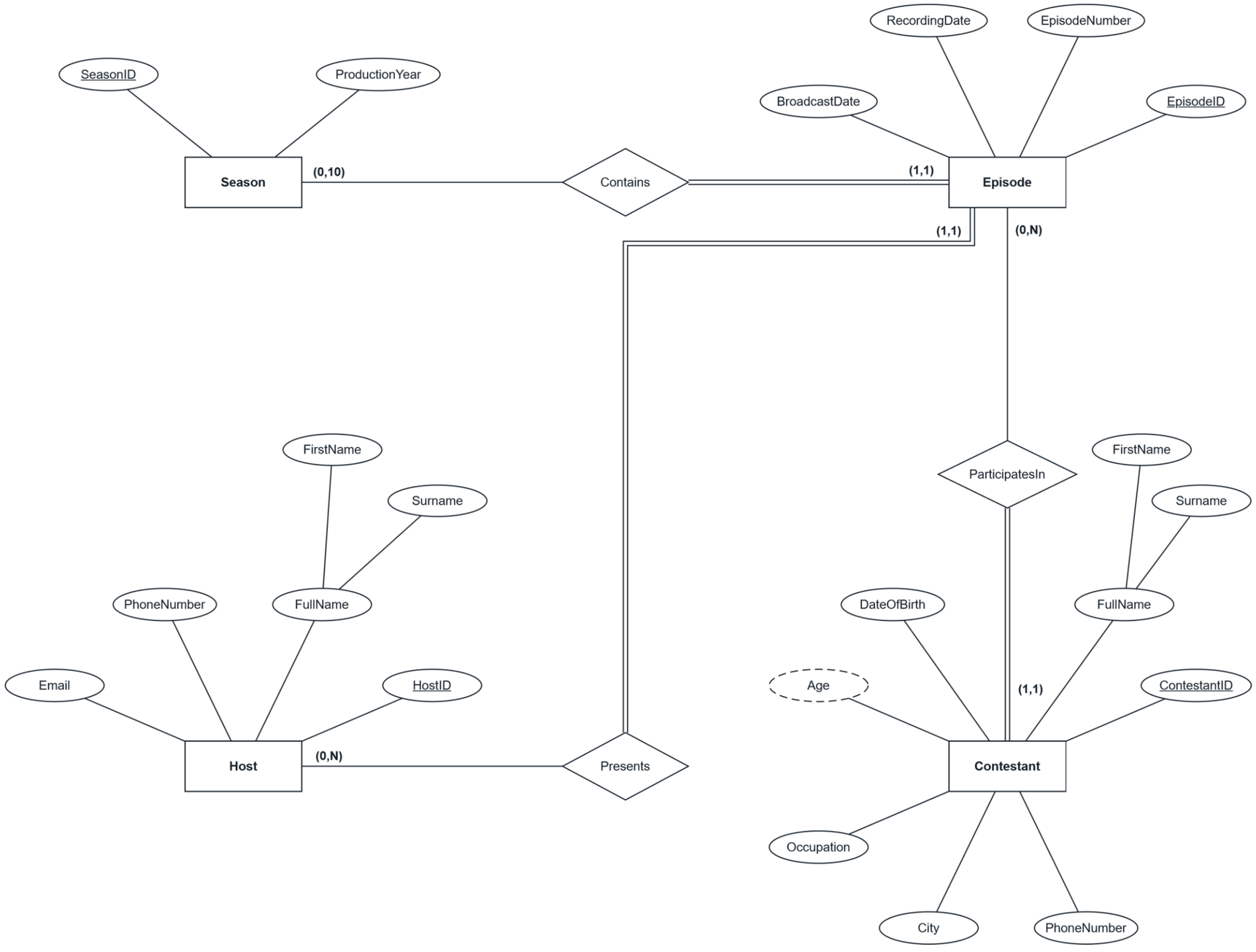
A2. The same question may be asked to the same contestant more than once. Each occurrence has a different AttemptNo. The brief does not rule out repeated questions.

A3. A joker type may be used more than once, as long as the contestant uses no more than four jokers in total. The brief lists three types without setting a separate limit for each type.

A4. Every recorded contestant has been asked at least one question. Contestant therefore has total participation in HasAttempt.

Use Case 1A - Production and participation

Chen ERD | Entity attributes continue in View 1B where marked. All relationships are binary.



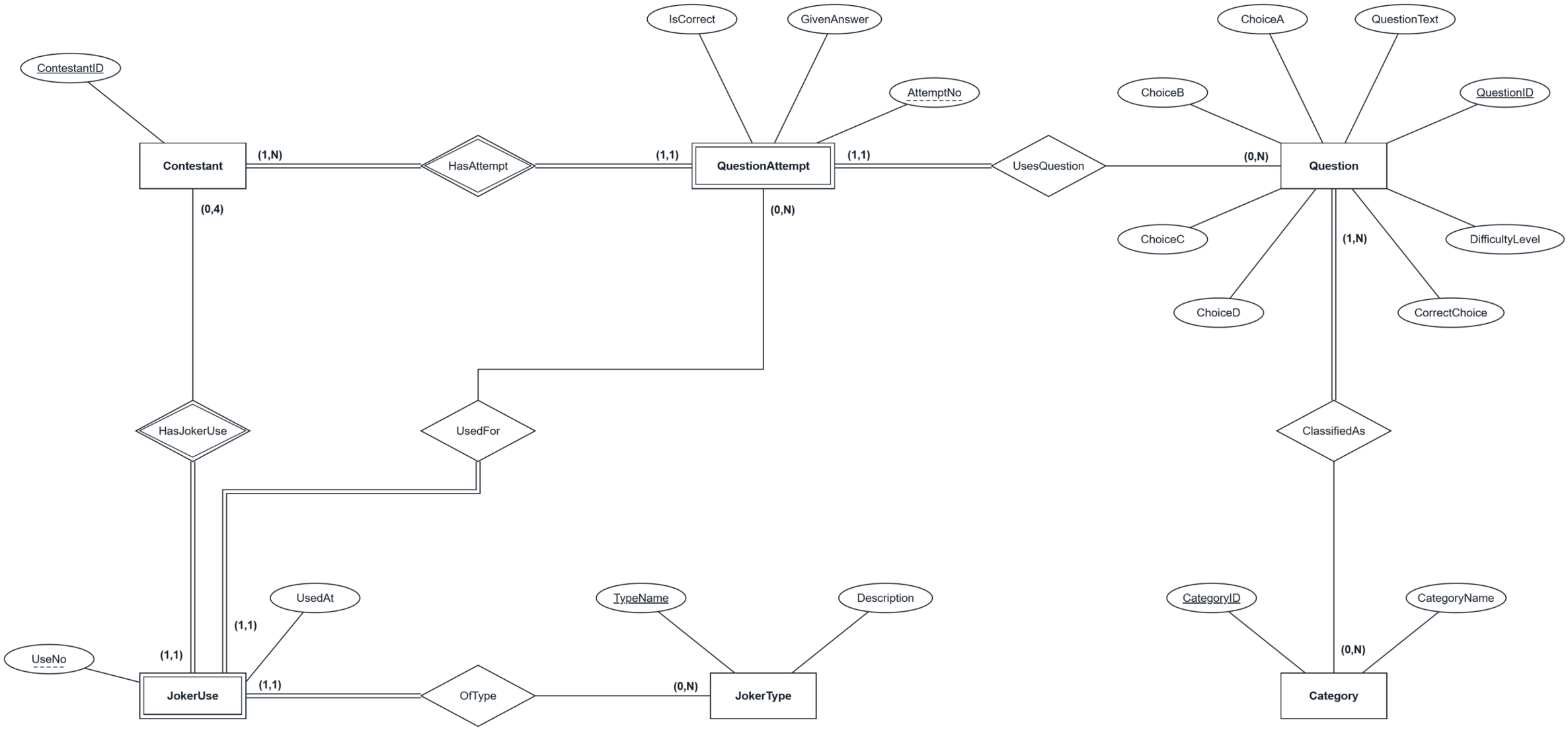
Notes

Contestant.Age is the number of completed years since DateOfBirth. Host.Email is unique in addition to HostID.

One Contestant record is kept for each person. It belongs to one episode, so a person can take part in the show only once.

Use Case 1B - Questions, answers and jokers

Chen ERD | Contestant is the same entity as in View 1A. Dashed underlines denote partial keys.



Notes

The full keys are (ContestantID, AttemptNo) for QuestionAttempt and (ContestantID, UseNo) for JokerUse. AttemptNo gives the question order.
A JokerUse must refer to a QuestionAttempt owned by the same contestant. IsCorrect is stored and checked against GivenAnswer = Question.CorrectChoice.
JokerType.TypeName can be Fifty-Fifty, Phone a Friend or Ask the Audience.

Use Case 2

CleanCampus

CleanCampus has laundry machines shared by dormitories. The model records these allocations, student stays and machine usage, along with maintenance contracts.

Entity identities

Entity	Chosen full key
LaundryRoom	RoomID
Machine	MachineID
MachineModel	ModelID
Dormitory	DormID
Student	StudentID
Stay	StayID
UsageSession (weak)	(MachineID, SessionNo)
Supplier	SupplierID
MaintenanceContract	ContractID

Design decisions

MachineModel holds the specifications shared by machines of that model. Capacity consists of MinLoad and MaxLoad; SupportedPrograms is multivalued. A machine has one model, while Brand remains an attribute of Machine, as in the brief.

AllocatedTo is a binary many-to-many relationship between machines and dormitories. UsageRank and FreeCreditAmount belong to each Machine-Dormitory pair, and IsLeadingDorm is derived from that pair's UsageRank.

We give each period of residence a StayID and link it to one Student and one Dormitory. A student who returns to a dormitory gets a new Stay record. CurrentlyIn shows the student's current dormitory and must agree with the open Stay.

UsageSession is weak under Machine, with SessionNo as its partial key and an identifying relationship to Machine. MaintenanceContract has its own identifier because the same supplier and machine can have more than one contract, including renewals.

Use Case 2 - Rules and assumptions

Business rules and additional constraints

1. Machine.Type is standard or industrial. Dormitory.Level is undergraduate or graduate. Student.Username is unique in addition to StudentID. Student.Age is stored because the brief gives no birth date.

2. Each machine has exactly one model and is in exactly one laundry room. MinLoad must not exceed MaxLoad. Supported program names are recorded in the model.

3. There is at most one allocation per Machine-Dormitory pair. It holds that pair's UsageRank and FreeCreditAmount. IsLeadingDorm is true exactly when UsageRank = 1 for that machine.

4. Each student has one current dormitory through CurrentlyIn and at least one Stay. Exactly one Stay is open, with EndDate = NULL, and its dormitory must match CurrentlyIn. A return to the same dormitory creates a new StayID. Stay periods for the same student must not overlap.

5. SessionNo is unique within a machine. Status is scheduled, completed, cancelled or interrupted. Duration is derived from StartTime and EndTime when both are known.

6. Completed and interrupted sessions must identify one dormitory, and the machine must be allocated to that dormitory at the time of use. A scheduled or cancelled session may have no actual-use dormitory.

7. Each maintenance contract is for one supplier and one machine and records StartDate, EndDate and ContractAmount. Renewals get separate contract records, even when the supplier and machine are unchanged.

Assumptions

A1. A machine is installed in one shared laundry room at a time, identified by RoomID. The model does not keep a history of room changes.

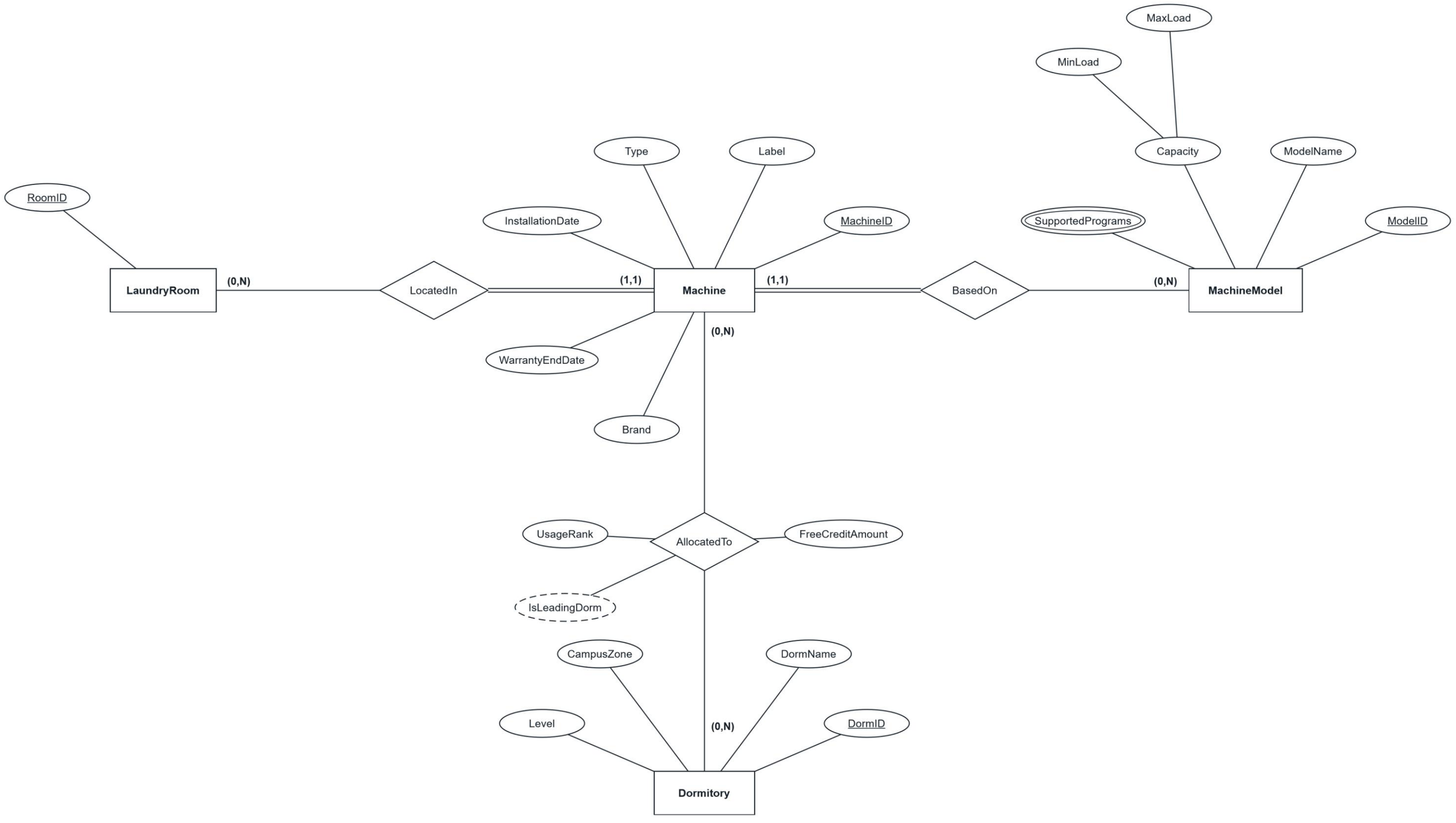
A2. Stay history includes the current open Stay. When a student leaves, EndDate is filled in and CurrentlyIn is kept consistent with the Stay record.

A3. UsageRank values are distinct positive integers for each machine. Rank 1 marks its single leading dormitory.

A4. A usage session ends on its SessionDate. EndTime may be unknown if a session is unfinished or cancelled. Duration is calculated only when both StartTime and EndTime are available.

Use Case 2A - Equipment and dormitory allocation

Chen ERD | Allocation attributes belong to the Machine-Dormitory pair.



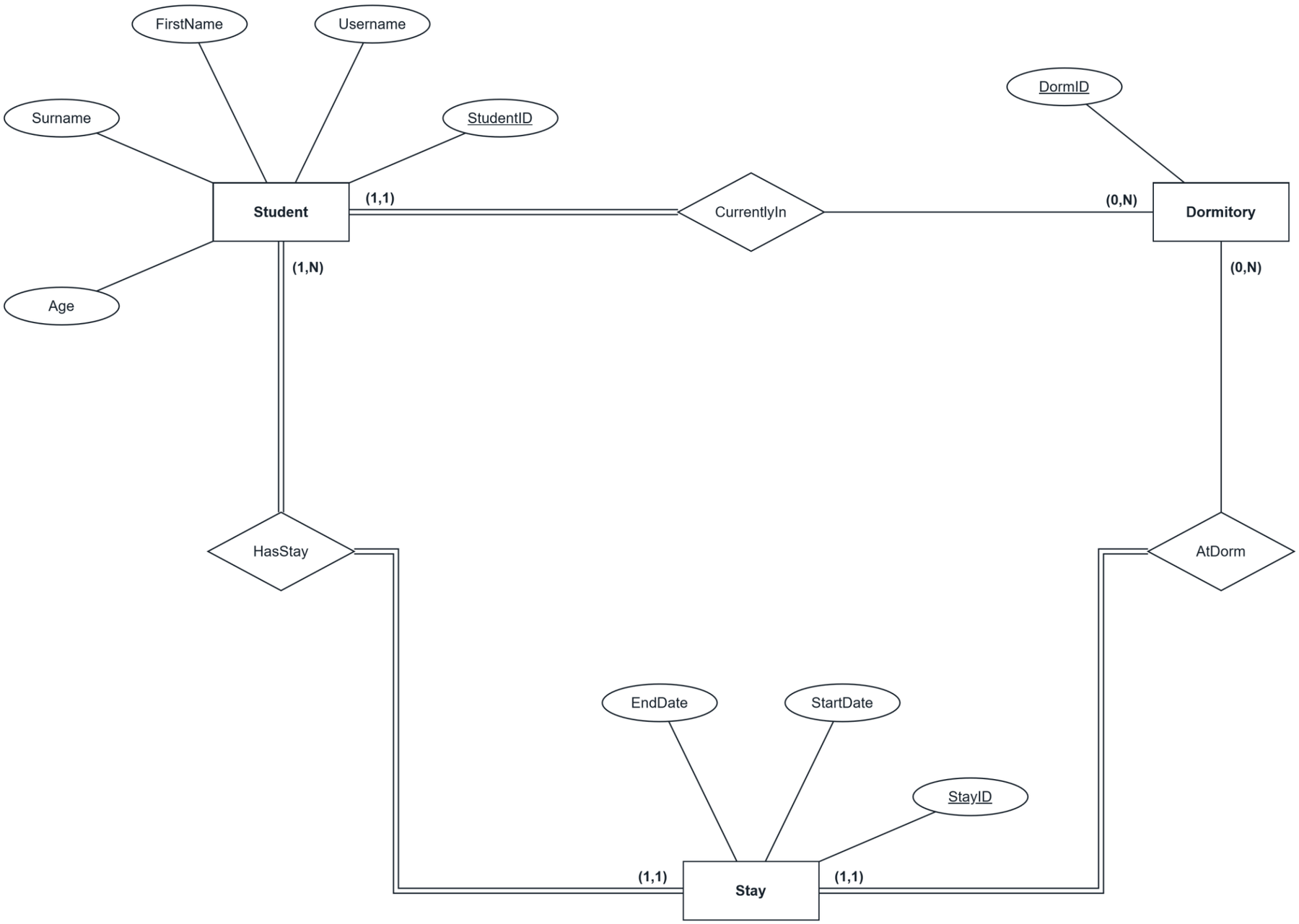
Notes

IsLeadingDorm is derived separately for each machine and is true exactly when UsageRank = 1.

Capacity is composite, with MinLoad and MaxLoad as its components. SupportedPrograms is multivalued.

Use Case 2B - Current residence and stay history

Chen ERD | Dormitory is the same entity as in View 2A. Each separate stay has its own StayID.



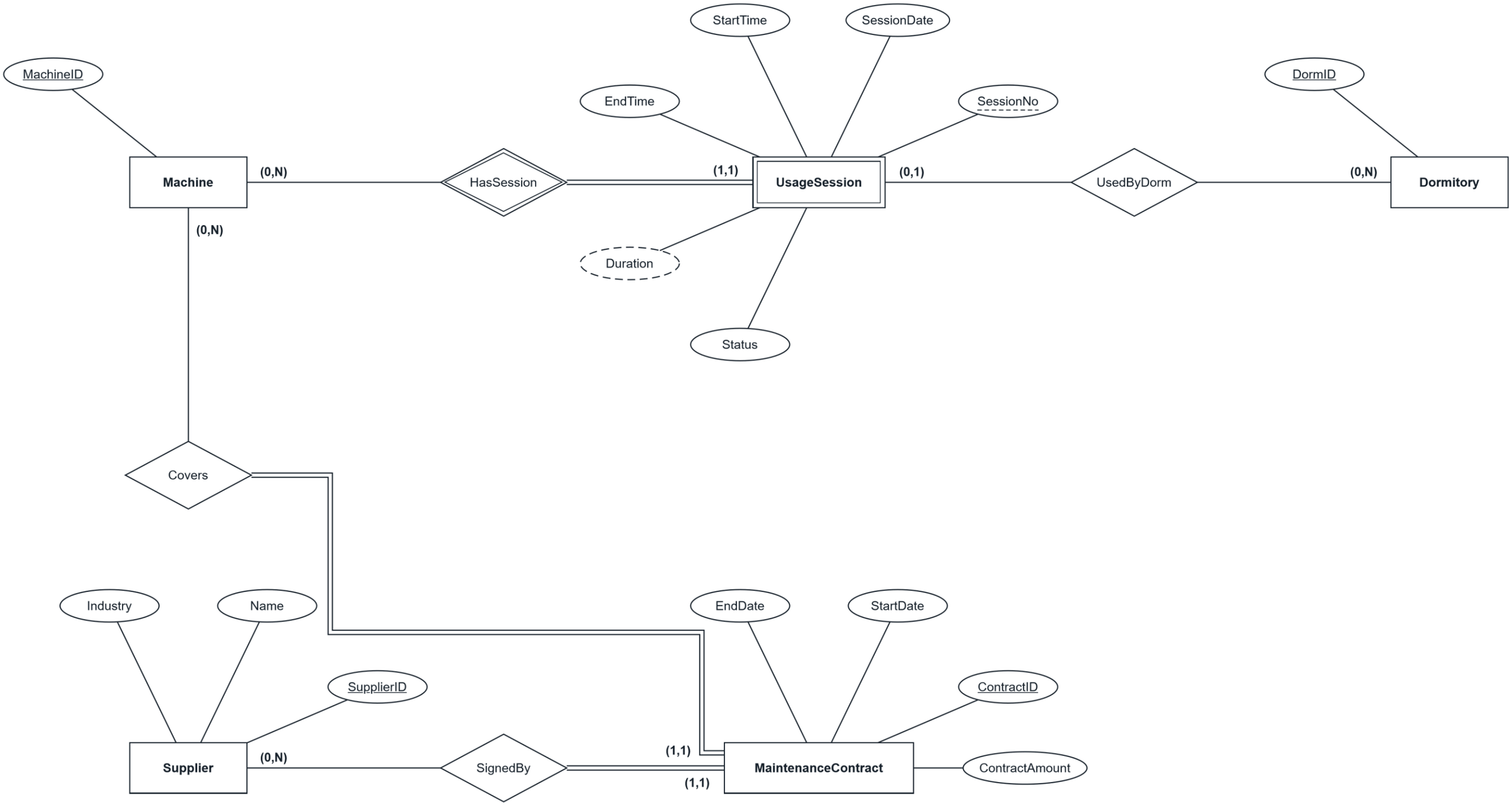
Notes

Student.Username is unique on its own, in addition to StudentID. Student.Age is stored because no birth date is given.

Each student has one open Stay with EndDate = NULL, and its dormitory must match CurrentlyIn. A return to the same dormitory gets a new StayID.

Use Case 2C - Usage sessions and maintenance

Chen ERD | Machine and Dormitory are references to View 2A. A usage session is weak under its machine.



Notes

UsageSession has the full key (MachineID, SessionNo). Duration is derived from StartTime and EndTime once both are available.
Completed and interrupted sessions must identify a dormitory. The machine must be allocated to that dormitory at the time of use.
A new maintenance agreement gets its own ContractID, even when the supplier and machine are the same.

Use Case 3

METU NCC KALTEV

The KALTEV model includes companies, their current staff and projects, office rentals, internships, meeting-room bookings and investments. Rentals and investments have separate records so that earlier transactions are retained.

Entity identities

Entity	Chosen full key
Company	TaxNumber
Employee	NationalID
Office	OfficeNumber
Rental	RentalID
Project	ProjectCode
Deliverable (weak)	(ProjectCode, DeliverableNo)
Intern	StudentID
MeetingRoom	RoomNumber
Booking (weak)	(RoomNumber, BookingDate, StartTime)
Investor	InvestorID
Investment	InvestmentID

Design decisions

A company can rent an office again or receive more than one investment. Rental and Investment therefore have their own identifiers. Each Investment is a dated transaction; these records are used to calculate Company.TotalInvestmentReceived and totals for an investor-company pair.

DeliverableNo is local to a Project. For Booking, the partial key (BookingDate, StartTime) is local to a MeetingRoom. Both weak entities have identifying relationships to their owners.

Company.Contact consists of Email and PhoneNumber. Employee.PhoneNumbers can hold several values, and CompanyAge is derived from FoundingDate. EmployeeCount* is stored even though it is calculated: it counts current WorksFor links and is updated automatically when those links change. The asterisk marks this storage requirement.

Use Case 3 - Rules and assumptions

Business rules and additional constraints

1. Company is identified by TaxNumber, and Employee by NationalID. Employee.Email must also be unique. Each employee currently works for one company through WorksFor, which records EmploymentStartDate.

2. Each company has one representative. An employee may represent at most one company and must currently work for that company. Represents is a subset of WorksFor for the same employee-company pair. This also means every company has at least one employee.

3. Each Rental links one company to one office and records StartDate, EndDate and MonthlyRent. A rental is ongoing when StartDate is on or before today and EndDate is NULL or later than today. Every company currently operating in KALTEV has at least one ongoing rental. Ended rentals are kept as history; an office may have no ongoing rental. Rental periods for the same office must not overlap.

4. A project belongs to one company. Its Status is proposed, active, completed or cancelled, and it has at least one Deliverable. Deliverable numbering starts at D1 for each project. ActualSubmissionDate may be NULL until the deliverable is submitted.

5. WorksOn records HoursPerWeek for each employee-project pair. Every project has one manager who also works on it, so Manages is a subset of WorksOn.

6. An intern belongs to one company through InternsAt and has one supervisor. The supervisor must work for the same company through WorksFor. Only current internships are recorded; the Intern record is removed when the internship ends.

7. There are exactly ten MeetingRoom records. Each booking is for one room and one company. Its full key is (RoomNumber, BookingDate, StartTime), so two bookings for the same room cannot start at the same time on the same date. Bookings in different rooms can start together.

8. An Investment has one investor and one receiving company. Company.TotalInvestmentReceived is the sum of Amount for that company's investments, or zero if it has none. To find the total from investor i to company c, we sum the investments linked to both i and c.

Assumptions

A1. WorksFor records current employment only. Previous employers are outside the model's scope. Employees work only on projects run by their current company.

A2. Rental periods are half-open intervals [StartDate, EndDate). A NULL EndDate means the interval is open-ended. A company may rent the same office again with a new RentalID.

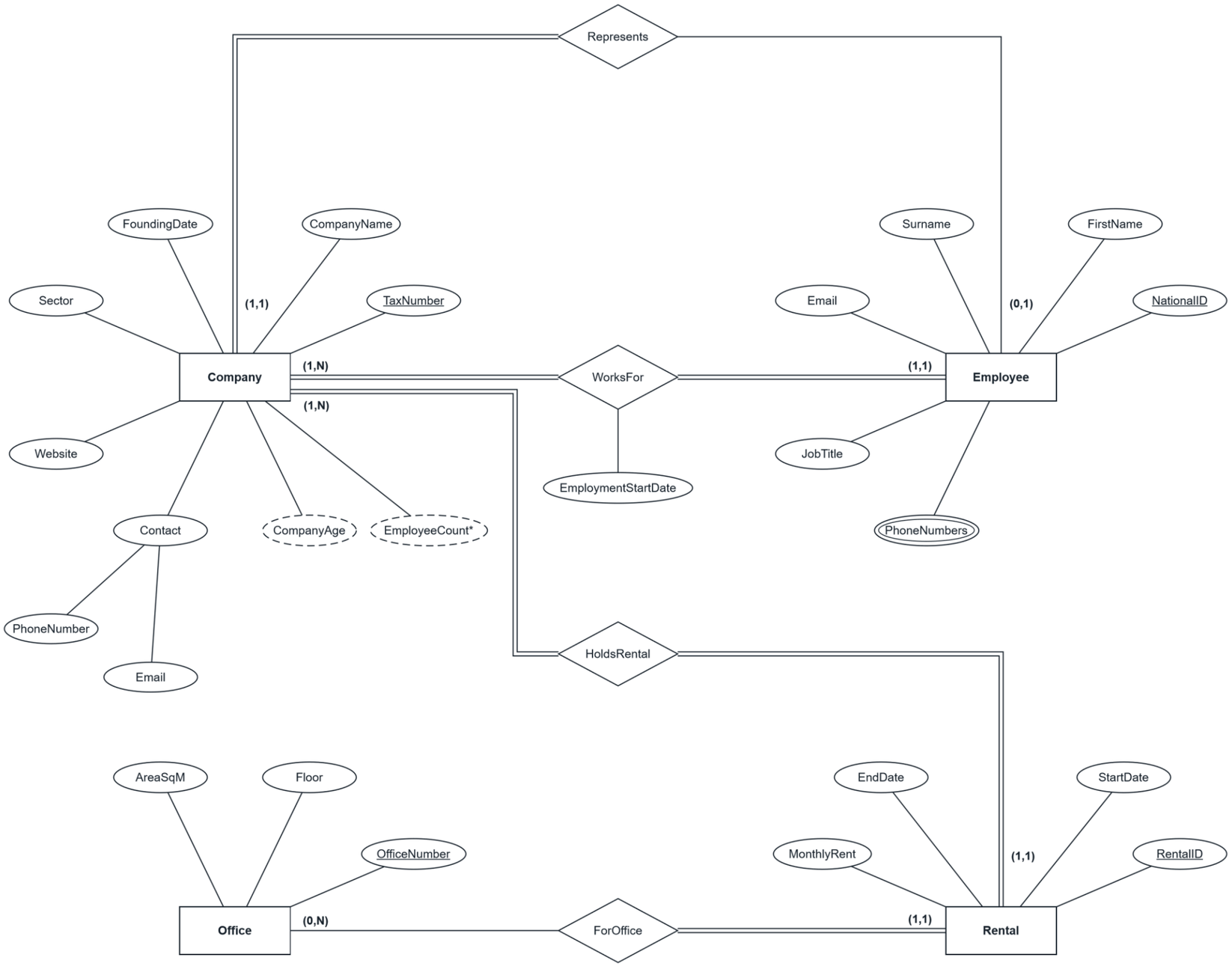
A3. A booking starts and ends on the same date, with StartTime earlier than EndTime. The key (RoomNumber, BookingDate, StartTime) prevents two bookings for the same room from starting on the same date at the same time.

A4. An investor may invest in a company more than once; each dated transaction has a separate InvestmentID. Investor email is not assumed to be unique. An investor may be registered before making an investment, which gives Investor partial participation in ProvidedBy.

A5. A company stays in the system after leaving KALTEV, with its past rentals still linked to it. Only companies currently operating in the building are required to have an ongoing rental.

Use Case 3A - Companies, employees and offices

Chen ERD | Company.EmployeeCount* is a stored, automatically maintained derived value.



Notes

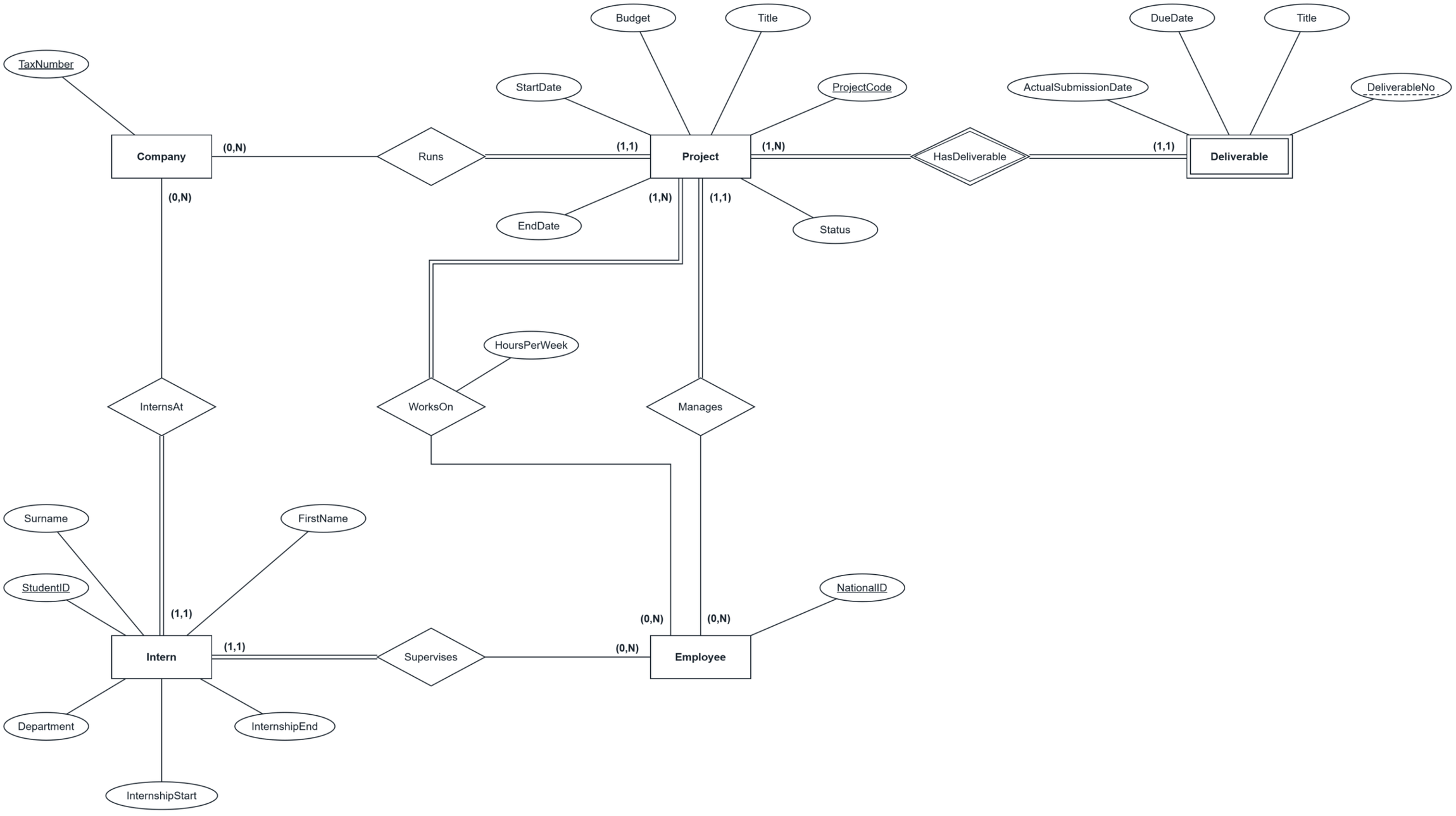
A company's representative must be one of its own current employees. Represents is therefore a subset of WorksFor.

Every company currently operating in KALTEV needs at least one ongoing rental. Rental periods for the same office must not overlap.

Employee.Email is unique in addition to NationalID. EmployeeCount* is the stored count of current employees in WorksFor and is updated automatically.

Use Case 3B - Projects, deliverables and interns

Chen ERD | Company and Employee refer to View 3A. Every project manager must also work on that project.

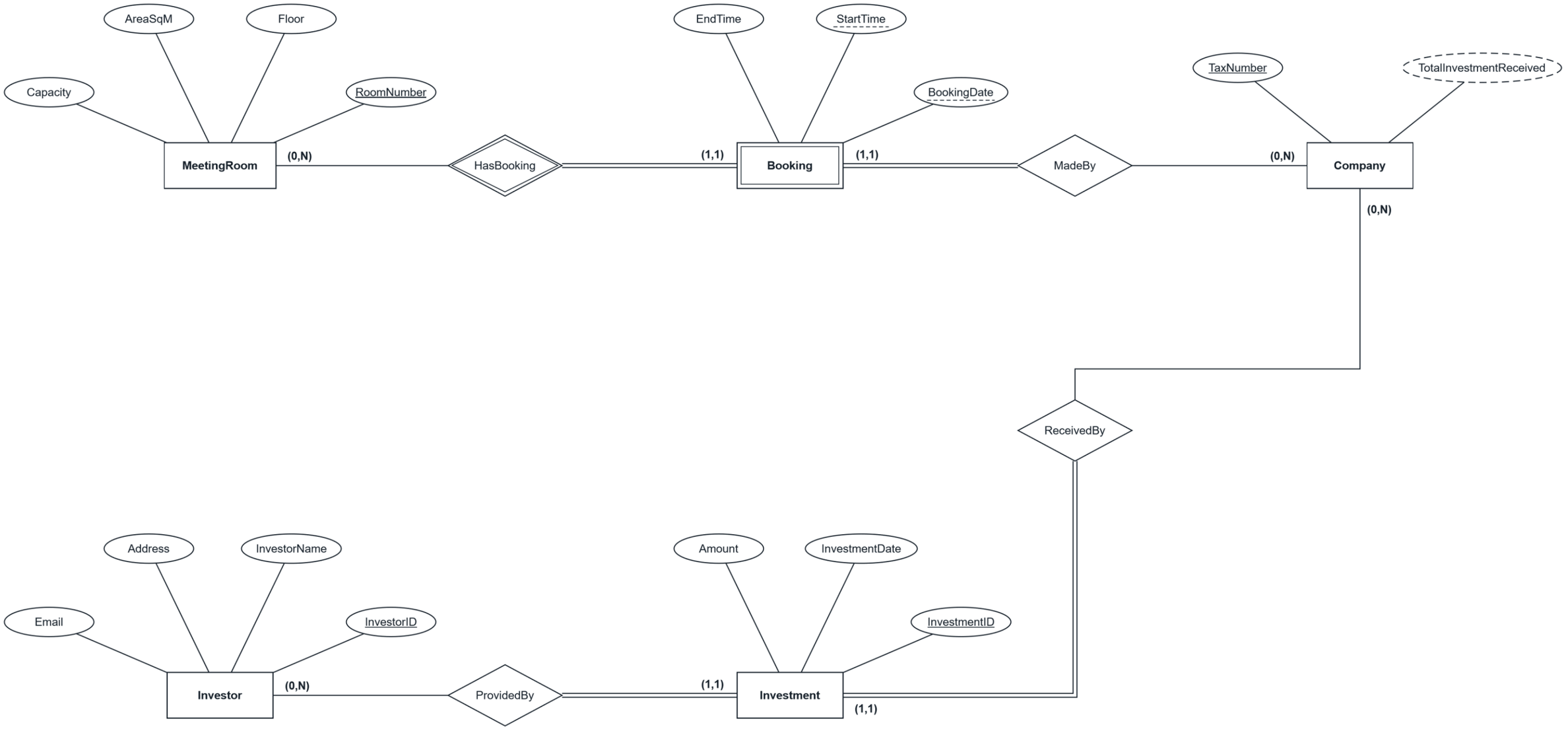


Notes

The full key for Deliverable is (ProjectCode, DeliverableNo). Numbering starts at D1 again for each project.
Manages is a subset of WorksOn: a project manager must also work on that project. All project workers belong to the company running it.
An intern and their supervisor must work for the same company. Only current internships are kept.

Use Case 3C - Meeting rooms and investment

Chen ERD | Company is the same entity as in View 3A; TotalInvestmentReceived is shown here.



Notes

Booking is identified by (RoomNumber, BookingDate, StartTime). Its partial key is (BookingDate, StartTime).

There are exactly ten MeetingRoom records. Each booking links one room to one company.

Each investment is kept as a separate transaction. Totals sum Amount for either a company or an investor-company pair.